

TradeLog — Project Proposal

A private trading journal for serious retail investors.

Capstone Project Proposal · Ben Rubinovitz · 2026

1. Summary

TradeLog is a full-stack web application that lets a small, trusted group of investors record, analyse, and reflect on their trades across **stocks, crypto, and forex** — in one place, with privacy by default and optional sharing.

Most retail traders today track their activity in a tangle of brokerage statements, spreadsheets, and screenshots. That works for *remembering* what they did, but not for *learning* from it. TradeLog turns a scattered trade history into a structured, queryable journal with real portfolio analytics — the kind of feedback loop that separates disciplined investors from gamblers.

2. The Problem

Retail trading has exploded, but the tooling hasn't kept up with how people actually trade:

- **Brokerages are siloed.** A trader with positions on a stock broker, a crypto exchange, and a forex platform has *three* histories and *zero* unified view.
- **Spreadsheets don't scale.** Manual logs are error-prone, can't compute risk-adjusted returns, and never tell you *why* a strategy is working.
- **Performance is misleading.** A naïve "account went up 10%" ignores deposits and withdrawals. Without **time-weighted** and **money-weighted** returns, you can't tell skill from cash flow.
- **There's no honest feedback loop.** Without notes tied to each trade, traders repeat the same mistakes — and forget the reasoning behind their wins.

The core pain is simple: **traders can see what they own, but not how well they trade.**

3. The Solution

TradeLog is a private trading diary with the analytical depth of a portfolio tool. Planned capabilities:

Capability	What it does
Unified trade log	One journal for stocks, crypto, and forex — long or short, open or closed.
Position tracking	Groups individual trade legs into positions, with averaged cost and realised P&L.
Honest performance	Time-weighted & money-weighted returns so deposits never masquerade as gains.
Analytics & charts	Equity curve, sector heatmap, allocation breakdown, and top movers.
Market data	Live (delayed) prices and in-trade price charts, fetched securely server-side.
Watchlist	Track symbols you don't own yet, with optional target-price alerts.
Reflection built in	Notes, tags, and screenshots attached to every trade — the learning layer.
Scenario playground	"What-if" and dollar-cost-averaging simulators to test ideas risk-free.
Privacy by default	Everything private unless you explicitly choose to share a trade with the group.

The guiding principle: **P&L and returns are always computed, never hand-typed.** The journal is a source of truth, not a place to fudge the numbers.

4. Who It's For

- **Primary:** A close-knit group of active retail traders (5–15 friends) who trade across multiple asset classes and want to improve, not just record.
- **Secondary:** Any disciplined individual investor who has outgrown spreadsheets and wants real analytics without handing their full financial history to a large commercial platform.

It is explicitly **not** a brokerage, a social network, or a get-rich-quick signal service. There is no trade execution and no payment logic — TradeLog is a **journal and analysis tool**, and that focus is a feature.

5. Why It Matters

- **Educational value.** A structured journal with notes and tags converts raw trade history into a personal dataset traders can actually learn from.
 - **Financial honesty.** Proper return math (TWR/MWR) gives an unflinching picture of performance — the single most useful thing a self-directed investor can have.
 - **Privacy as a stance.** Built for a trusted group, private by default. Users share on their own terms rather than broadcasting by default.
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6. Technical Approach (*overview*)

A modern, type-safe full-stack web app, deployed on managed infrastructure:

- **Frontend & framework:** Next.js (App Router) with TypeScript and a dark, responsive UI built in Tailwind CSS.
- **Backend:** Server actions and API routes — **no client-side database access**, keeping all data logic and secrets on the server.
- **Database:** PostgreSQL, accessed through the Prisma ORM with versioned migrations.
- **Authentication:** Google OAuth, so there are no passwords to manage or leak.
- **Market data:** A pluggable provider layer (e.g. Finnhub / CoinGecko) called **only from the server**, with caching so the app stays responsive and never crashes on stale or missing prices.
- **Visualisation:** Charting libraries for equity curves, allocation views, and in-trade price charts.

Key design commitments: **money values stored as precise decimals, dates in UTC and shown in the user's timezone**, and a strict rule that **financial figures are derived, never manually entered**.

7. Project Plan (*high-level milestones*)

1. **Foundations** — Authentication, trade CRUD, positions, cash flows, and the core data model.
2. **Analytics & visualisation** — Equity curve, sector/allocation charts, dividend tracking, and live market data.
3. **Playground** — What-if and dollar-cost-averaging simulators.
4. **Polish & sharing** — Optional trade sharing, watchlist alerts, and UX refinement.

Each milestone is independently demonstrable, so progress is visible at every stage rather than only at the end.

8. Success Criteria

The project will be considered successful if it can:

- Log and group trades across all three asset classes with **computed** P&L.
 - Report **time- and money-weighted returns** that correctly exclude deposits and withdrawals.
 - Render an accurate equity curve and allocation breakdown from real trade data.
 - Fetch and display live market data **without ever exposing API keys to the browser**.
 - Run entirely on free-tier managed infrastructure, demonstrating a deployable, real-world architecture.
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9. Scope & Constraints

In scope: journalling, analytics, market-data display, simulation, and opt-in sharing within a trusted group.

Out of scope (by design): trade execution, brokerage integration, payments or subscriptions, and any form of public social feed. These boundaries keep the project focused and achievable within a capstone timeframe.

TradeLog is a tool I'd want to use myself — and that, more than anything, is why I want to build it.